

Integration of artificial intelligence in the digital preservation of academic repositories and scientific data in higher education libraries

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Abstract

Purpose – This paper aims to investigate the transformative role of artificial intelligence (AI) in digital preservation in higher education libraries (HEL) with a focus on integrating advanced technologies to overcome traditional challenges in knowledge management. The study presents a systematic analysis that encompasses process automation, metadata creation and enrichment and the protection of scientific research data. Initially, it demonstrates how natural language processing (NLP)-based tools enhance the automatic extraction of essential information, optimizing operational efficiency and ensuring the integrity of repositories. Then, it discusses the importance of metadata standardization – through international standards such as Dublin Core and PREMIS – to ensure interoperability between systems and promote global collaboration. The paper also highlights the relevance of AI in mitigating risks, such as format obsolescence and data corruption and in democratizing access to information, expanding digital inclusion. Finally, future directions are discussed that emphasize the need for modular tools, collaborative governance and robust ethical policies to consolidate the sustainability and adaptability of digital preservation solutions.

Design/methodology/approach – A systematic literature review was conducted in Web of Science and Scopus (2020–2025), using four iterative search expressions and predefined inclusion/exclusion criteria (e.g. open access, “information science library science”). After screening, 48 articles were included for qualitative, interpretative synthesis under a constructivist paradigm. The review maps AI applications across preservation workflows and standards (e.g. Dublin Core, Preservation Metadata Maintenance Activity – PREMIS, Metadata Encoding and Transmission Standard – METS).

Findings – The review identifies five consistent themes: automation of curation tasks via natural language processing (NLP) for scalable, more consistent metadata; semantic enrichment to improve discovery and reuse; predictive analytics for format obsolescence and proactive migration; interoperability gains through standards-driven metadata; and governance needs around transparency, bias mitigation and capacity-building. Benefits co-exist with constraints related to infrastructure, skills and ethical safeguards.

Research limitations/implications – Results reflect the chosen timeframe, databases and filters and are limited to published, open-access literature in Library Information Science venues. No primary testing of tools was performed. Future work should include comparative case studies, shared evaluation metrics and longitudinal assessments of AI-enabled preservation.

Practical implications – Academic libraries can prioritize AI policies and governance, invest in modular and standards-aligned tools, develop staff competencies (AI literacy) and embed ethical review (transparency, audits, bias mitigation) to scale preservation, improve interoperability and free staff time for higher-value tasks.

Originality/value – However, as pointed out by (Subaveerapandiyar and Ugwulebo, 2024), barriers related to technological infrastructure and financial support still limit the ability of many institutions, especially in less developed contexts, to integrate advanced AI solutions. These limitations highlight the importance of investments in professional training and in the creation of adaptive tools that can be adjusted to the needs and constraints of each context. Despite the advances achieved, the application of AI in academic repositories is not without challenges, with ethical issues being a central concern. Transparency in the algorithms used for data curation is essential to avoid bias and ensure that technologies are inclusive and equitable. Hansson and Dahlgren (2022) warn of the risks of exclusion of underrepresented groups and the need for ethical guidelines that ensure equity in access to and use of scientific information. Furthermore, Sikhakhane and Mthombeni



(2024) explore the legal and psychological implications of the use of AI in academic libraries, suggesting that clear regulatory frameworks are essential to protect the interests of users and ensure the accountability of institutions that implement these technologies. Issues, such as the protection of personal data and information security, also emerge as critical challenges, requiring a multidisciplinary approach to address the complexities associated with AI governance. Sustainable governance is also a crucial aspect in integrating AI into academic repositories. (Strecker *et al.* 2023) argue that long-term data preservation requires collaborative governance models that prioritize infrastructure resilience and adaptability to technological change. This view is complemented by Azeroual (2021), who reinforces the importance of data quality as an essential pillar for the ongoing integrity of digital collections. In this context, Bossaller and Million (2023) highlight the challenges associated with data management and the need to integrate modern solutions without compromising the reliability of information already stored. The future of AI in academic repositories presents several promising directions for future research. One priority area is the development of modular and adaptable AI tools that can be customized according to institutional and regional contexts. Bamgbose *et al.* (2024) suggest that these solutions can democratize the use of AI, enabling institutions with limited resources to implement advanced digital preservation practices. This approach could be explored in future research that evaluates how these modular tools can be designed to maximize their impact in different contexts. Another promising direction is the construction of global networks of academic repositories, which facilitate interoperability and promote the sharing of good practices. Moyo and Bangani (2023) argue that international partnerships are essential to connect repositories in a global ecosystem, expanding the reach and relevance of collections. Furthermore, Frank and Wylie (2023) highlight the role of semantic enrichment as a facilitator for interdisciplinary connections, allowing data stored in academic repositories to be used more broadly and impactfully. Finally, ethics must remain at the center of discussions about AI in academic repositories. Shal *et al.* (2024) emphasize that organizational leaders play a crucial role in creating policies that promote transparency, inclusion and social responsibility. Melero *et al.* (2023) complement this perspective by pointing out that the dissemination of scientific knowledge must be accompanied by practices that ensure equity in access to and use of information. The ethical and sustainable integration of AI is essential to ensure that the benefits of these technologies are widely distributed and that their impact is positive for science and society as a whole. AI has already proven to be an indispensable tool for transforming academic repositories.

Keywords Artificial intelligence, Digital preservation, Academic repositories, Higher education libraries, Scientific data

Paper type Literature review

Introduction

Recent studies indicate that although higher education libraries (HELs) are critical to preserving scientific knowledge, traditional digital curation practices face significant challenges. For example, Blumesberger (2020) and Yoon *et al.* (2022) demonstrate that exponential data growth and rapid technological obsolescence compromise the integrity of repositories. Given this finding, the integration of solutions based on artificial intelligence (AI) is proposed to automate processes and reduce errors, aligning procedures with international standards, as advocated by Chassanoff and Altman (2020) and Murillo and Yoon (2024).

Digital preservation involves more than ensuring the longevity of archives and repositories; it is about ensuring that scientific knowledge remains accessible, relevant and usable over time, even in the face of technological innovations and changes in storage standards. According to Blumesberger (2020), initiatives that use AI, such as PHAIDRA, have demonstrated potential to harmonize data from different institutions, promoting interdisciplinary collaboration and increasing the global relevance of academic repositories. These technological advances, however, come with new ethical, operational and technical challenges that need to be resolved in an integrated manner.

In light of this, this paper aims to explore how AI can be applied in HEL to address three fundamental challenges. First, the ability of AI to identify and mitigate risks in the preservation of academic repositories will be investigated. Machine learning tools have proven effective in

predicting risks, such as format obsolescence and data corruption, enabling the development of proactive and robust preservation strategies (Yoon *et al.*, 2022; Frank and Wylie, 2023). Second, this work examines how AI can automate and enrich metadata creation, overcoming the limitations of manual methods and ensuring interoperability between academic systems and repositories. Third, the automated generation of consistent metadata, aligned with international standards such as *Dublin Core* and *PREMIS*, is essential to enhance global collaboration and accessibility (Cox, 2023; Monteagudo-Haro and Prieto-Gutierrez, 2024).

This article is organized to address, in sequence, the main aspects of the application of artificial intelligence in digital preservation in academic environments. Initially, the section Implementation of Artificial Intelligence Technologies in the Preservation of Digital Academic Repositories presents the current context and challenges, emphasizing how AI can transform traditional curation practices. Next, the section Implementation of Artificial Intelligence for the Automation of Metadata in Academic Repositories explores solutions for the automated creation and standardization of metadata, aiming to increase interoperability. Section 3, Application of Artificial Intelligence in the Preservation of Scientific Research Data, investigates strategies to protect and preserve scientific data sets, ensuring their integrity and long-term accessibility. Finally, the section Development of Advanced Strategies for Integration and Sustainability in Digital Data Preservation proposes models and guidelines for collaborative and sustainable governance, justifying the need for policies that guarantee evolution and inclusion in the context of digital preservation.

In this sense, this study will analyze the role of AI in the protection and preservation of scientific research data sets, with a focus on classification, compression and long-term accessibility, ensuring that data is protected against obsolescence and loss (Moyo and Bangani, 2023; Ngulube and Mosha, 2024).

The potential impact of these approaches is significant. For example, Adika and Kwanya (2020) and Huang *et al.* (2021) highlight how automation, driven by AI technologies, has alleviated the manual workload in HEL, allowing professionals to focus their efforts on strategic activities, such as data policy management and long-term preservation. Furthermore, the interoperability promoted by metadata standardization generates substantial benefits, connecting local repositories to global knowledge networks and fostering more effective collaborations.

However, the implementation of AI in HEL is not without challenges. Ethical issues, such as algorithmic biases and digital exclusion, raise significant concerns about equity in access to knowledge. Hansson (2020) warns of the risks associated with the lack of transparency in algorithms, while Sikhakhane and Mthombeni (2024) emphasize the need for clear regulatory frameworks to protect users' interests and ensure that technological solutions are inclusive and accountable. Furthermore, challenges related to sustainable governance and the resilience of technological infrastructures are crucial to ensuring that digital preservation initiatives are effective in the long term.

This study contributes to the existing literature by proposing an integrative model that explores the potential of AI as a strategic tool for the transformation of digital preservation in HEL. The analysis is based on the review of studies, highlighting both the opportunities and challenges associated with the implementation of advanced technologies in academic repositories. By positioning AI as a catalyst for innovations in the field of digital preservation, this work seeks to provide theoretical and practical *insights* that can inform future research and guide institutional policies aimed at sustainability and inclusion.

Finally, this paper argues that balancing technological innovation with ethical principles will be essential to ensure that the benefits of AI are widely distributed and that its impact is positive for both science and society. By exploring practical and sustainable approaches, this

research lays a solid foundation for future advances, fostering a future in which academic knowledge is accessible, relevant and impactful for all global communities.

Methodology

This study follows a systematic literature review design, using qualitative, interpretative analysis to synthesize findings (constructivist paradigm). According to [Carrillo Ramos and Santiesteban Naranjo \(2014\)](#), constructivism emphasizes the understanding of phenomena based on the interpretation of the subjects involved and is an appropriate approach for studies that explore complex processes, such as the application of AI in digital preservation. This paradigm rejects the view of knowledge as something fixed and objective, favoring a contextualized analysis, in which interpretations emerge from the active involvement of the researcher with the object of study.

To conduct this research within this epistemological framework, a qualitative approach was adopted, which allows for an in-depth analysis of the studies, emphasizing the meaning of the data rather than its quantification. As pointed out by [Coutinho \(2023\)](#), qualitative research seeks to understand emerging processes and patterns and is widely used in studies in the social and human sciences, where data are analyzed in an interpretative way. The qualitative approach is particularly relevant when investigating complex and multifaceted issues, such as the intersection between AI and digital preservation in HEL, as it allows us to explore the subjectivity of the processes involved and identify nuances that could be ignored in purely quantitative analyses. The progressive development of the four search expressions employed in this systematic review is synthesized in [Table 1](#), providing a clear overview of their conceptual evolution and refinement.

When analyzing the four search expressions, we observed a clear evolution in line with the expectations of this research. We identified common results in the first three expressions, while we varied the words according to their relevance. In the quantitative results, there was an increase in the first two expressions and a decrease in the third, due to the inclusion of new terms (marked in green) and the exclusion of others (marked in red). This refinement process aimed to reach the ideal expression (the fourth expression).

It is important to mention that the year 2025 was included to reflect possible updates and changes. A pattern observed in the first three expressions was the absence of results in SCOPUS. The discrepancy between the Web of Science and SCOPUS in the retrieval of results became evident; for example, in the second and third expressions, SCOPUS frequently confused “AI” with “air”, questioning the accuracy of the terms used, while in the first expression, it indicated inadequacy of the words.

Finally, in the fourth search expression used for this investigation, we noticed a recovery of results in SCOPUS and a greater restriction of the results. This adjustment was crucial to ensure the relevance and contextual relevance of this investigation, finding a more efficient way to meet the proposed objectives. This refinement instills confidence that the development of the four search expressions will lead us to a rigorous and quality study.

The selection of these databases is based on the need to guarantee the quality and relevance of the retrieved studies, as recommended by [Vilelas \(2022\)](#), who highlights the importance of using reliable and methodologically robust sources for the construction of scientific knowledge.

Next, a *screening process* was conducted, in which strict inclusion and exclusion criteria were applied to further refine the set of studies analyzed. This stage followed the recommended methodological guidelines for systematic reviews and allowed us to identify the most relevant studies, ensuring that only those that directly addressed the relationship between AI, digital preservation and HEL were incorporated into the final analysis. This

Table 1. Development and evolution of search expressions (from the first to the one used in research)

Expressions	Research data	Web of Science	Scopus
First search expression	("research data" OR "open research data" OR "research data services" OR "research data management" OR "data literacy" AND "artificial intelligence" AND "academic librar *" OR "research librar *" OR "university librar *" AND "digital preservation" AND "study case" AND "systematic review")	21,208 results Time period (2020–2024 + 2025) – 9659 results Document type (<i>review article</i>) – 1,102 results	0 results
Second search expression	("research data" OR "open research data" OR "research data services" OR "research data management" OR "data literacy" AND "artificial intelligence" AND "academic librar *" OR "research librar *" OR "university librar *" AND "digital preservation" AND "institutional repository" OR "research repository")	21,337 results Timeline (2020–2024 + 2025) – 9,733 results Document type (<i>review article</i>) – 1,111 results	0 results
Third search expression	("research data" OR "open research data" OR "research data services" OR "research data management" OR "data literacy" AND "artificial intelligence" OR "AI" AND "academic librar *" OR "research librar *" OR "university librar*" and "digital preservation" AND "institutional repository" OR "research repository")	21,444 results Time period (2020–2024 + 2025) – 9,840 results Document type (<i>review article</i>) – 1,119 results	0 results
Fourth search expression	("research data" OR "open research data" OR "research data management" OR "data literacy" AND "artificial intelligence" OR "AI" AND "academic librar *" OR "research librar *" AND "digital preservation" OR "digital archiving" AND "institutional repository")	21,117 results Timeline (2020–2024 + 2025) + Document type (<i>article</i>) + Open access + Research area (information science, library science) - 274 results	37 results Timeline (2020–2024 + 2025) + Document type (<i>article</i>) - 13 results

Source(s): Own elaboration

selection process guaranteed the quality and relevance of the data analyzed, reinforcing the validity of the results obtained.

Thus, the combination of the constructivist paradigm, the qualitative approach and a rigorous process of research and source selection ensured the methodological robustness of this study. The use of recognized databases, the careful application of search and screening strategies, as well as the automated verification of duplications and manual selection of articles, ensured the scientific validity of the research. Thus, this research not only offers an in-depth analysis of the impact of AI on digital preservation in HEL but also establishes a rigorous and replicable methodology for future research in the area.

After the screening process, 48 articles met all criteria and were included in the review for in-depth analysis.

Implementation of artificial intelligence technologies in the preservation of digital academic repositories

The digital preservation of academic repositories represents a critical challenge in the contemporary scenario. With the exponential growth in the volume of scientific data, institutions face increasing difficulties in ensuring the integrity and accessibility of their collections. The obsolescence of digital formats, the lack of standardization and the inadequacy of technological infrastructures are problems that threaten the longevity of information essential to science and society. In this context, AI emerges as a transformative tool, capable of redefining preservation strategies and fostering innovative solutions. [Table 2](#) summarizes the key stages involved in the implementation of AI-driven technologies for the digital preservation of academic repositories, highlighting the associated workflows and decision points.

Digital preservation is at the center of contemporary academic and technological discussions, given the increasing reliance on digital technologies to store and disseminate information. The exponential growth of academic data and the diversification of digital formats have made it imperative to adopt innovative and scalable practices for information management.

One of the biggest challenges in digital preservation is detecting obsolete formats, a task that traditionally requires manual work and in-depth technical knowledge. AI offers a scalable solution to this issue through machine learning algorithms trained to identify patterns in file formats and automatically classify those at risk of obsolescence. As discussed by [Yoon et al. \(2022\)](#), AI-based curation *frameworks* can perform this analysis in real time, significantly reducing manual efforts and enabling faster corrective actions. Tools like these help institutions anticipate problems and plan interventions efficiently. [Figure 1](#) contrasts manual and AI-automated methods, illustrating their respective performance metrics and operational characteristics.

Table 2. Preservation process with AI

Process steps	Description	Reference
Obsolete format detection	Identifying file formats that are becoming obsolete	Yoon et al. (2022)
Automatic classification	AI algorithms automatically classify files at risk	Harisanty et al. (2024)
Predictive obsolescence analysis	Predictive models use historical data to predict format obsolescence	Frank and Wylie (2023)
Corrective action	Implementing measures to migrate or preserve formats at risk	Yoon et al. (2022)
Intervention planning	Proactive planning to avoid future obsolescence	Chiarelli et al. (2022)

Source(s): Own elaboration

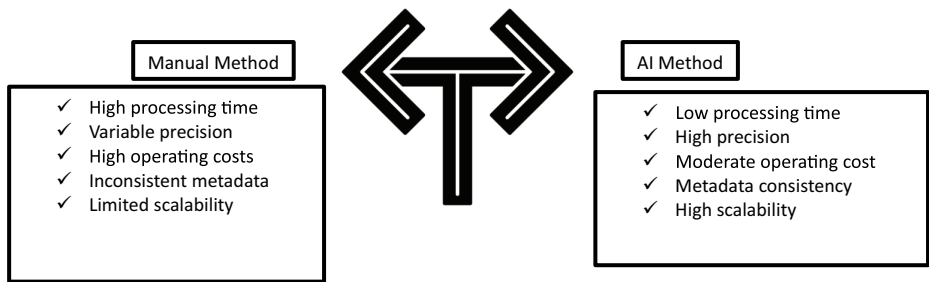


Figure 1. Comparison of manual methods vs automation with AI

Source: Prepared based on [Enwald *et al.* \(2022\)](#); [Cox \(2023\)](#); [Chigwada \(2024\)](#); [Harisanty *et al.* \(2024\)](#); [Murillo and Yoon \(2024\)](#)

In addition to detection, predicting the lifespan of digital formats is a promising application of AI. Predictive models based on deep learning use historical data and trend analysis to determine the optimal time for format migration. [Frank and Wylie \(2023\)](#) highlight that incorporating risk analysis into these models increases their accuracy and usefulness, allowing managers to prioritize resources for critical formats. This approach not only protects data but also improves the financial and operational efficiency of institutions that deal with large volumes of information. [Figure 2](#) depicts the proposed system architecture integrating academic repositories, metadata infrastructures and scientific data sets into a cohesive digital preservation framework.

Standardization and interoperability are essential elements for digital preservation in an increasingly interconnected world. AI-based recommendation systems can suggest file formats that are compatible with global standards, facilitating collaboration between institutions and ensuring long-term accessibility. This is a critical point in regions with limited technological infrastructure. [Subaveerapandiyan and Ugwulebo \(2024\)](#) demonstrate

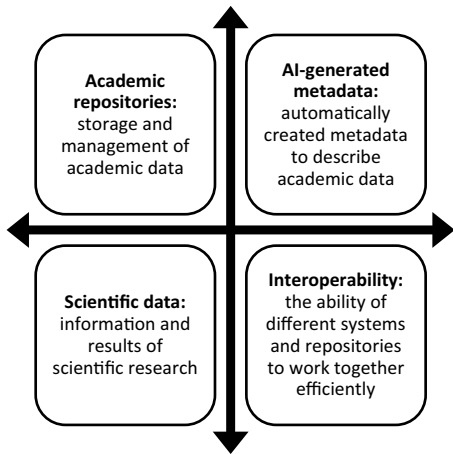


Figure 2. System architecture for the integration of academic repositories, metadata and scientific data

Source: Prepared based on [Blumesberger \(2020\)](#); [Enwald *et al.* \(2022\)](#); [Chigwada \(2024\)](#); [Murillo and Yoon \(2024\)](#)

how AI tools can be adapted to meet the needs of repositories in challenging contexts, promoting equity in access to scientific knowledge. Additionally, as discussed by [Enwald et al. \(2022\)](#), standardization of metadata can significantly improve data interoperability and accessibility, facilitating collaboration between different academic institutions.

The impact of AI on digital preservation goes beyond technical improvements. It represents a paradigm shift, enabling repositories to adopt a proactive, data-driven approach. This transformation not only protects the global scientific memory but also ensures that it remains accessible for future generations. However, the implementation of these technologies requires attention to ethical issues, such as the potential increase in digital inequalities and the need to ensure that AI tools are designed in an inclusive and sustainable way. As highlighted by [Cox \(2023\)](#), adopting ethical practices in the development of AI technologies is essential to ensure equity and inclusion.

The digital preservation of academic repositories goes beyond a mere technical challenge and becomes a global responsibility. As we move toward a future in which digital information becomes the main vector of knowledge, the integration of AI becomes essential. This scenario imposes the need to reflect on how to ensure that technological advances not only preserve, but also expand access to knowledge. As pointed out by [Yoon et al. \(2022\)](#), curation *frameworks* can incorporate collaborative practices to ensure data sustainability and accessibility. Furthermore, [Frank and Wylie \(2023\)](#) emphasize that proactive approaches, supported by AI-based risk analysis, are crucial to address the ethical and technical challenges of digital preservation, promoting solutions that are ethical and inclusive.

Metadata automation in academic repositories

Metadata generation, which was previously feasible, becomes impractical in the face of the monumental amount of data generated continuously. Metadata automation with the application of AI and NLP emerges as a crucial solution, providing efficiency, accuracy and interoperability while meeting international standards.

The exponential growth of academic data poses challenges for manual curation, which cannot keep up with the scale and complexity of today's needs. [Yoon et al. \(2022\)](#) state that manual curation often results in inconsistencies and difficulties in ensuring interoperability between systems, compromising access to and preservation of academic knowledge. In this context, metadata automation, using AI-based tools, presents itself as an indispensable solution to maintain the integrity and accessibility of academic repositories. A comparative analysis of the challenges of manual curation versus AI-assisted solutions is presented in [Table 3](#), outlining the relative advantages and limitations of each approach.

Among the most promising technologies, NLP stands out as a sub-area of AI that analyzes and interprets human texts. According to [Murillo and Yoon \(2024\)](#), NLP algorithms allow the automated extraction of essential information, such as titles, authors, keywords and abstracts. This approach speeds up the creation of metadata, significantly reduces human errors and ensures greater consistency. Automation eliminates limitations associated with manual validation, as automatically generated metadata can be adjusted to meet international standards, such as *Dublin Core* and *PREMIS*.

It is worth mentioning that [Monteagudo-Haro and Prieto-Gutierrez \(2024\)](#) highlight that the adoption of global standards in Spanish repositories not only improved accessibility but also increased the international relevance of these collections.

Metadata automation with AI enables semantic enrichment, connecting concepts and aggregating contextual information that improves data retrieval and access. [Frank and Wylie \(2023\)](#) highlight that semantic enrichment improves search accuracy and promotes interoperability between different systems, which is essential for interdisciplinary repositories.

Table 3. Challenges in manual curation vs. AI solutions

Curation stage	Challenges in manual curation	AI solutions	Reference
Data collection	Data volume becomes excessive	Automated extraction	Chassanoff and Altman (2020) ; Yoon et al. (2022)
Analysis	Lack of standardization	NLP for semantic and syntactic analysis	Murillo and Yoon (2024)
Classification	High time and inconsistency	Automated categorization algorithms	Chigwada (2024)
Storage	Inefficiency in data organization	Standards-based organization such as <i>Dublin Core</i>	Blumesberger (2020) ; Subaveerapandiyan and Ugwulebo (2024)

Source(s): Own elaboration

By integrating data from different areas of knowledge, AI facilitates connections that expand the reach and applicability of stored knowledge. [Figure 3](#) illustrates how semantic enrichment can be applied across heterogeneous data types to enhance discoverability and interoperability.

Thus, the semantic enrichment of metadata is highlighted by [Frank and Wylie \(2023\)](#) as a critical factor in increasing the relevance of academic collections. This practice allows repositories not only to store data, but also to promote the collection of information in interdisciplinary contexts.

Compliance with international standards is another critical aspect of metadata automation. [Subaveerapandiyan and Ugwulebo \(2024\)](#) discuss how AI-based tools ensure adherence to global standards, promoting uniformity in data organization and preservation. This is especially important for academic repositories seeking to integrate their collections into global networks, ensuring that their information remains accessible and interoperable in the long term. [Figure 4](#) visualizes the degree of compliance with international standards achieved by AI-based digital preservation tools.

Metadata automation provides benefits that go beyond operational efficiency. [Strecker et al. \(2023\)](#) emphasize that reducing human error and increasing scalability are fundamental to the sustainability of digital repositories. Automated systems ensure that metadata meet

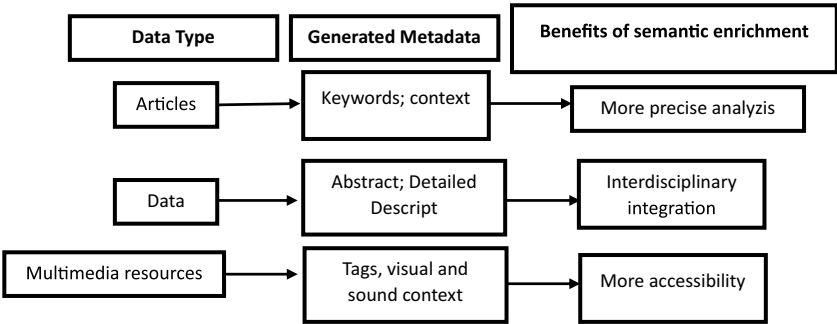


Figure 3. Semantic enrichment applied to different types of data

Source: Prepared based on [Blumesberger \(2020\)](#); [Corrêa and de Sousa \(2022\)](#); [Hansson and Dahlgren \(2022\)](#); [Frank and Wylie \(2023\)](#); [Murillo and Yoon \(2024\)](#)

International standard	Objective	AI contribution
•Dublin Core •PREMIS •METS	•Organization •Preservation •Interoperability	•Generating accurate descriptions •Data integration validation •Conformity in formats for exchange

Figure 4. Compliance with International Standards

Source: Prepared based on [Blumesberger \(2020\)](#); [Chassanoff and Altman \(2020\)](#); [Yoon et al. \(2022\)](#); [Cox \(2023\)](#); [Subaveerapandiyan and Ugwulebo \(2024\)](#)

consistent quality standards, expanding institutions’ capacity to handle growing volumes of information. This transformation prepares academic repositories for a future in which digital information will be the main vector of scientific knowledge.

Furthermore, data quality and curation are essential prerequisites for open research. [Azeroual \(2021\)](#) highlights that metadata automation must ensure high data quality from the outset, which is crucial for data integrity and trust in the information provided.

Repositories such as PHAIDRA at the University of Vienna already demonstrate how automation can be effectively used to manage research data comprehensively ([Blumesberger, 2020](#)). The integration of archival practices with AI technologies is also crucial to ensure long-term preservation ([Corrêa and de Sousa, 2022](#)). [Figure 5](#) presents the PHAIDRA repository as an illustrative case study of metadata automation and AI-supported digital preservation strategies.

Metadata automation should be considered an essential component in the digital preservation of academic repositories. By integrating advanced technologies such as NLP and promoting semantic enrichment, AI not only addresses contemporary challenges but also sets a standard for the preservation and access to knowledge on a global scale. This approach

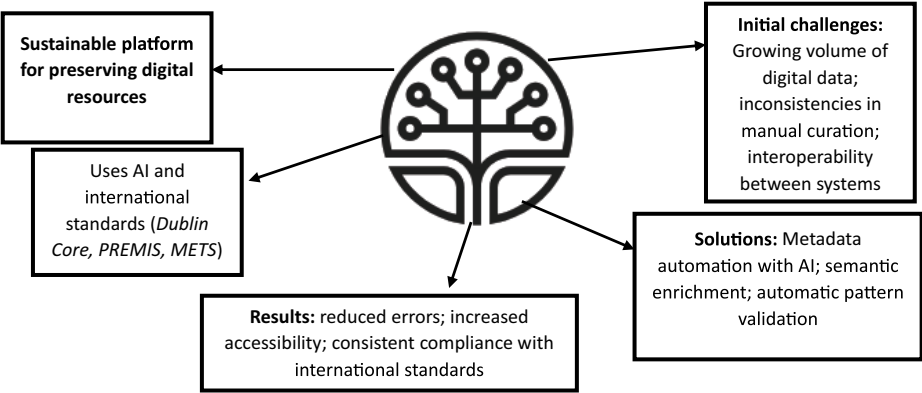


Figure 5. PHAIDRA repository

Source: Prepared based on [Blumesberger \(2020\)](#)

ensures that academic repositories remain relevant, accessible and interoperable, cementing AI as an indispensable tool for the future of science.

Application of artificial intelligence in the preservation of scientific research data

The preservation of scientific research data presents unique challenges, characterized by the diversity of formats and the sensitivity of the information. With the increasing production of scientific data, institutions face the complex task of ensuring the integrity, accessibility and security of this data over time. In this context, the application of AI technologies offers innovative solutions that can transform the management and preservation of scientific research data. Research data are generated in a wide variety of formats and have varying levels of sensitivity, from personal information of study participants to confidential data from scientific experiments. The diversity and complexity of this data require advanced approaches to its management and preservation.

As highlighted by [Chiarelli et al. \(2022\)](#), preserving research data requires implementing strategies that can handle the heterogeneity of formats and ensure long-term accessibility. One of the most promising applications of AI in preserving research data is the automatic classification of *data sets*. AI-based tools can automatically classify data by type, format and use, making it easier to organize and manage this data. As noted by [Harisanty et al. \(2024\)](#), automatic classification not only reduces the time and effort required to manage data but also increases the accuracy and consistency of classification, ensuring that data is stored appropriately and accessibly. [Figure 6](#) outlines the workflow for the automated classification of scientific research data through machine learning techniques.

Predictive analytics is another powerful AI tool that can be applied to research data preservation. Using machine learning algorithms, predictive analytics can identify data formats that are at risk of becoming obsolete. This predictive capability allows institutions to take proactive steps to migrate data to more stable and durable formats before they become inaccessible. As discussed by [Frank and Wylie \(2023\)](#), integrating risk analytics into predictive models increases the accuracy of these predictions and allows for more efficient management of resources allocated to data preservation. [Figure 7](#) demonstrates the use of predictive analytics to anticipate format obsolescence risks within digital collections.

Implementing efficient compression and storage strategies is essential to cope with the growing volumes of research data. AI-based compression techniques can reduce file sizes without significant data loss, optimizing the use of storage space and reducing associated

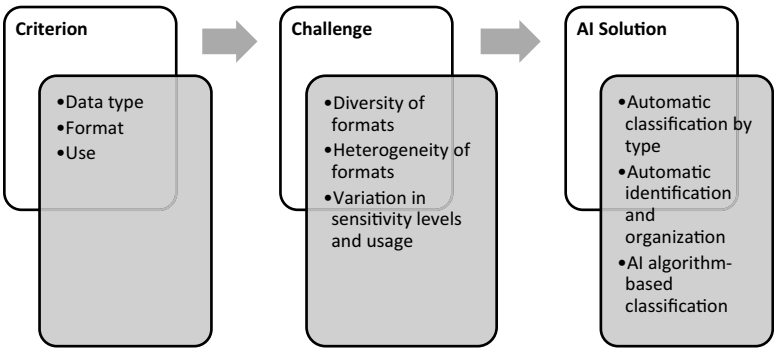


Figure 6. Automatic data classification

Source: Prepared based on [Chiarelli et al. \(2022\)](#); [Harisanty et al. \(2024\)](#); [Murillo and Yoon \(2024\)](#)

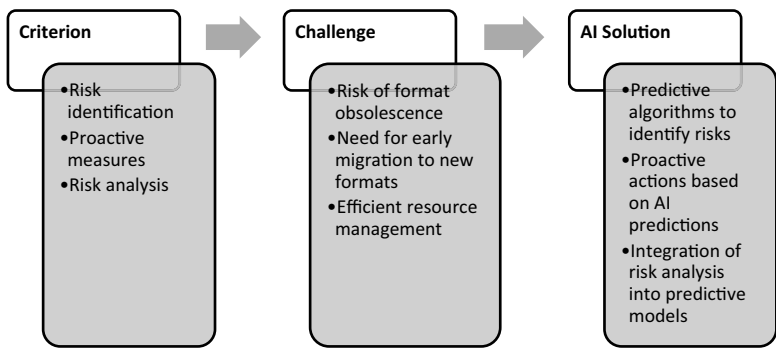


Figure 7. Predictive analysis of obsolescence risks

Source: Prepared based on [Chiarelli et al. \(2022\)](#); [Frank and Wylie \(2023\)](#); [Murillo and Yoon \(2024\)](#)

costs. As pointed out by [Moyo and Bangani \(2023\)](#), the adoption of efficient compression technologies is crucial to ensure the long-term sustainability of research data repositories. Furthermore, efficient storage strategies, such as the use of cloud-based solutions, can improve the accessibility and security of data, ensuring that it is available for future analysis and research. [Figure 8](#) summarizes AI-enabled strategies for optimizing data compression and storage efficiency in large-scale repositories.

The application of AI to the preservation of scientific research data brings a series of benefits that go beyond simple process automation. Key benefits include reduced human error, increased efficiency and improved data accessibility and security. Automating data classification and analysis processes significantly reduces the incidence of human error, ensuring that data is managed accurately and consistently.

As noted by [Kautonen \(2024\)](#), AI can eliminate many of the vulnerabilities associated with manual data management, producing more reliable and accurate results. The use of AI tools allows institutions to manage larger volumes of data more efficiently. This not only

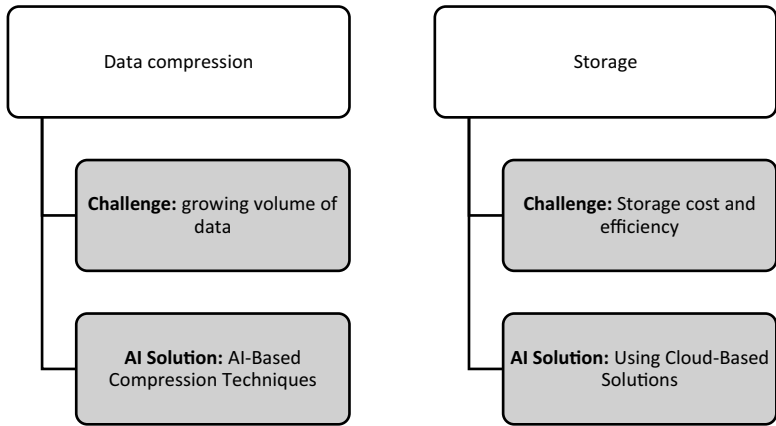


Figure 8. Efficient compression and storage strategies

Source: Prepared based on [Hansson and Dahlgren \(2022\)](#); [Moyo and Bangani \(2023\)](#); [Chigwada \(2024\)](#)

saves time and human resources for other important activities but also improves the ability of institutions to respond quickly to data preservation needs. According to [Strecker et al. \(2023\)](#), the adoption of advanced AI technologies can transform the way research data is managed, making the process more efficient and effective. Implementing efficient compression and storage strategies significantly improves the accessibility and security of research data. AI tools can ensure that data is stored securely and remains accessible for future analysis and research. According to [Chiarelli et al. \(2022\)](#), the adoption of AI technologies in research data management promotes the long-term sustainability and accessibility of data repositories.

The preservation of scientific research data faces unique challenges that require advanced and innovative solutions. The application of AI technologies to automatically classify *data sets*, predictively analyze obsolescence risks and implement efficient compression and storage strategies offers a robust approach to the management and preservation of this data. The benefits include reduced human error, increased efficiency and improved data accessibility and security, solidifying AI as an essential tool in the preservation of scientific research data. The adoption of these advanced technologies prepares academic and research institutions for a future where digital information will be the main vector of knowledge.

Development of advanced strategies for integration and sustainability in digital data preservation

Integrating academic repositories, metadata and scientific data into a single AI-based digital preservation strategy is essential to ensure the longevity, accessibility and usefulness of scientific data. This integration not only facilitates efficient data management but also promotes a holistic and sustainable approach to digital preservation. Connecting academic repositories with metadata and scientific data requires a robust and interoperable architecture that allows fluid communication between different systems. According to [Blumesberger \(2020\)](#), using repositories as tools for comprehensive research data management is essential to ensure long-term preservation and accessibility. AI integration can automate the creation and enrichment of metadata, ensuring that these repositories are consistent and accessible. AI tools, especially those based on NLP, can extract relevant information from academic texts and generate rich and consistent metadata. As discussed by [Chiarelli et al. \(2022\)](#), using machine learning models for metadata management not only speeds up the process but also ensures that metadata meets international interoperability standards such as *Dublin Core* and *PREMIS*. This automation is crucial for the efficient integration of repositories and the promotion of a cohesive digital preservation strategy. Standardization of metadata is another critical aspect for the integration of academic repositories. As pointed out by [Enwald et al. \(2022\)](#), the adoption of global metadata standards improves interoperability between systems and facilitates collaboration between institutions. AI can play a vital role in recommending compatible metadata formats and ensuring that data is accessible and usable in the long term. [Figure 9](#) highlights the main benefits of metadata standardization as a means to ensure long-term interoperability and data integrity.

The implementation of an AI-based digital preservation strategy should be guided by ethical and sustainable principles. Minimizing bias in AI models is essential to ensure equity and inclusion. As highlighted by [Cox \(2023\)](#), AI models can reflect and amplify biases present in training data. Therefore, it is crucial to implement bias mitigation practices, such as diversifying training data sets and conducting regular audits of models. Ensuring transparency in automated processes is another pillar of an ethical and sustainable approach. Transparency promotes user trust and facilitates institutional adoption of AI technologies. According to [Harisanty et al. \(2024\)](#), clear documentation of automated processes and

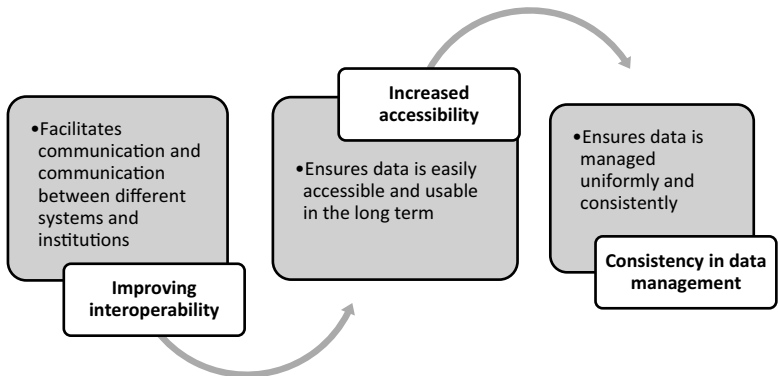


Figure 9. Benefits of metadata standardization

Source: Prepared based on [Blumesberger \(2020\)](#); [Chiarelli et al. \(2022\)](#); [Enwald et al. \(2022\)](#)

decisions is essential to ensure that users understand and trust AI systems. Furthermore, transparency in processes allows for the identification and correction of potential flaws or biases, promoting accountability and continuous improvement of systems. [Figure 10](#) presents recommended best practices for mitigating algorithmic bias and ensuring ethical governance in AI applications.

Thus, Hansson and Dahlgren (2022) warn of the risk of algorithmic biases, which can perpetuate inadvertent exclusions or discrimination. To mitigate these risks, the authors advocate transparency in algorithms and the adoption of robust ethical guidelines.

Promoting institutional and user acceptance is crucial to the success of an AI-based digital preservation strategy. The adoption of these technologies must be accompanied by capacity-building and training efforts to ensure that information professionals are prepared to use and manage AI tools. According to [Kautonen \(2024\)](#), building AI skills in academic libraries is essential to ensure successful and sustainable adoption. Furthermore, involving users and other *stakeholders* in the development and implementation of AI technologies can increase the acceptance and effective use of these tools. [Figure 11](#) provides an overview of strategies

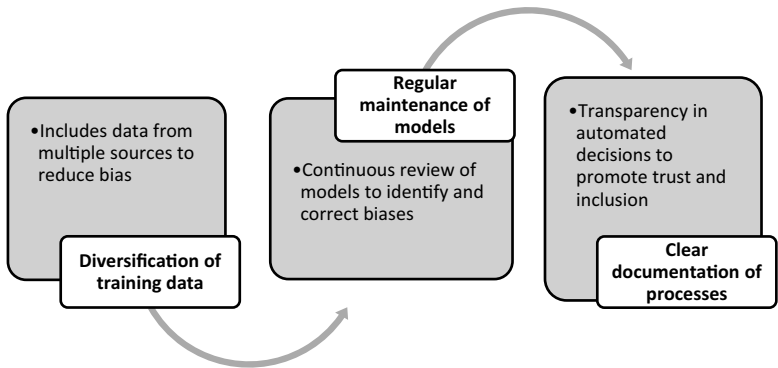


Figure 10. Bias mitigation practices in AI models

Source: Prepared based on [Cox \(2023\)](#); [Harisanty et al. \(2024\)](#)

to foster professional acceptance of AI and promote continuous capacity-building among information specialists.

Integrating academic repositories, metadata and scientific data into a single AI-based digital preservation strategy offers numerous benefits, including operational efficiency, improved data accessibility and the promotion of an ethical and sustainable approach. Operational efficiency is achieved through process automation and metadata standardization, enabling institutions to manage larger volumes of data with greater accuracy and consistency. According to [Murillo and Yoon \(2024\)](#), integrating AI into data curation improves the efficiency of digital preservation processes, freeing up time and resources for other essential activities.

Improved data accessibility is promoted by standardizing metadata and interoperability between systems, ensuring that data are available for future analysis and research. According to [Subaveerapandiyan and Ugwulebo \(2024\)](#), using AI tools to recommend compatible metadata formats facilitates collaboration between different institutions and promotes long-term accessibility. Promoting an ethical and sustainable approach is achieved by mitigating biases, ensuring transparency in automated processes and fostering institutional and user acceptance. [Hansson and Dahlgren \(2022\)](#) warn of the risk of algorithmic biases that can perpetuate inadvertent exclusions or discrimination, reinforcing the need for robust ethical guidelines. Furthermore, [Harisanty et al. \(2024\)](#) emphasize that the acceptance of AI technologies in libraries and academic institutions depends on both transparency in the processes and the empowerment of the professionals involved. These principles ensure that the implementation of AI technologies is equitable and inclusive, promoting trust and acceptance of the technologies by users. A promising direction in this context is the development of modular and adaptable tools that can be customized to meet the needs of different institutions. According to [Kautonen \(2024\)](#), building institutional competencies for the use of these tools is fundamental for their effective and sustainable adoption.

The integration of academic repositories, metadata and scientific data into a single AI-based digital preservation strategy is crucial to ensuring the longevity, accessibility and usefulness of scientific data. [Blumesberger \(2020\)](#) highlights that standardization and interoperability are

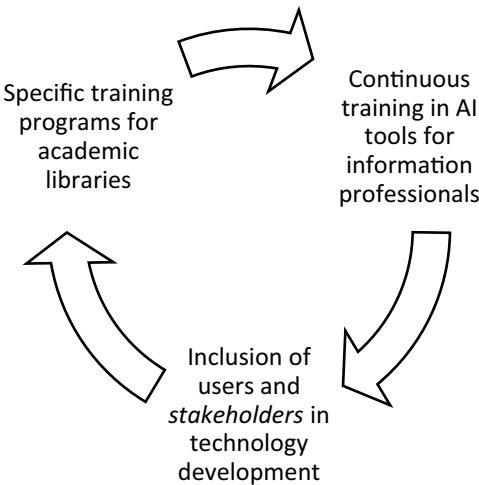


Figure 11. AI training and acceptance strategies
Source: Prepared based on [Harisanty et al. \(2024\)](#); [Kautonen \(2024\)](#)

essential elements to ensure the long-term preservation and global accessibility of academic repositories. Furthermore, [Cox \(2023\)](#) points out that transparency in AI models and regular audits are essential to avoid algorithmic biases and ensure the reliability of automated systems. In this sense, an ethical and sustainable approach, which includes minimizing bias in AI models, ensuring transparency in automated processes and promoting institutional and user acceptance, is essential for the success of this strategy. The benefits of this integration include operational efficiency, as evidenced by [Murillo and Yoon \(2024\)](#), who demonstrate how automation significantly improves metadata management in digital repositories. Improved data accessibility is another central benefit, promoted by the standardization of metadata and interoperability between systems. [Subaveerapandiyan and Ugwulebo \(2024\)](#) indicate that the adoption of AI tools facilitates collaboration between different institutions and promotes long-term accessibility. In this way, AI consolidates itself as an essential tool in digital preservation, as long as it is implemented responsibly and aligned with global governance standards, as advocated by [Sikhakhane and Mthombeni \(2024\)](#).

Findings

The integration of AI into academic repositories has changed the way scientific information is managed, preserved and shared. In a global context of exponential data production, traditional digital curation and preservation practices have proven insufficient to deal with the challenges posed by this growing volume of information. In this scenario, AI emerges as an essential tool, enabling significant advances in process automation, metadata quality and global interoperability of repositories. However, the advances achieved coexist with technical, ethical and structural challenges that require attention and provide directions for future research.

Among the main results achieved, the automation of processes related to data curation stands out. NLP-based technologies have enabled the automatic extraction and cataloging of relevant information, such as authors, keywords and abstracts, significantly reducing the manual workload. [Chassanoff and Altman \(2020\)](#) demonstrate that the use of NLP not only reduces human effort but also increases consistency and accuracy in the generation of metadata, factors essential for the integrity of academic repositories. This operational efficiency was corroborated by [Huang et al. \(2021\)](#), who observed how automation, implemented in Chinese university libraries, facilitated the scalability needed to manage the growing volumes of academic publications. By replacing manual practices, which are susceptible to errors and inconsistencies, automation allowed information professionals to direct their efforts to more strategic activities, such as data policy management and long-term preservation.

Metadata quality and standardization emerge as central outcomes of the application of AI in academic repositories. The generation of standardized metadata, following international standards such as *Dublin Core* and *PREMIS*, has been widely recognized as essential to ensure interoperability between systems and integration into global knowledge networks. [Cox \(2023\)](#) emphasizes that standardization facilitates the efficient sharing of data between institutions, promoting more effective and global collaborations. Practical examples include initiatives such as *PHAIDRA*, described by [Blumesberger \(2020\)](#), where AI is used to harmonize data from different institutions, promoting interdisciplinary collaboration and strengthening the relevance of academic collections. Similarly, [Monteagudo-Haro and Prieto-Gutierrez \(2024\)](#) report that Spanish university repositories have managed to expand their global reach by adopting open standards, connecting to an international ecosystem of repositories and researchers.

Another significant impact of AI on academic repositories is the democratization of access to information. Machine learning-based tools have enabled a greater reach of academic data, promoting inclusion and expanding global accessibility. [Ngulube and Mosha \(2024\)](#) highlight that inclusive and ethical practices are essential to ensure that the benefits of AI technologies are widely distributed, reaching both large research centers and academic communities in regions with more limited resources.

Discussion

However, as pointed out by [Subaveerapandiyan and Ugwulebo \(2024\)](#), barriers related to technological infrastructure and financial support still limit the ability of many institutions, especially in less developed contexts, to integrate advanced AI solutions. These limitations highlight the importance of investments in professional training and in the creation of adaptive tools that can be adjusted to the needs and constraints of each context.

Despite the advances achieved, the application of AI in academic repositories is not without challenges, with ethical issues being a central concern. Transparency in the algorithms used for data curation is essential to avoid bias and ensure that technologies are inclusive and equitable. [Hansson and Dahlgren \(2022\)](#) warn of the risks of exclusion of underrepresented groups and the need for ethical guidelines that ensure equity in access to and use of scientific information. Furthermore, [Sikhakhane and Mthombeni \(2024\)](#) explore the legal and psychological implications of the use of AI in academic libraries, suggesting that clear regulatory frameworks are essential to protect the interests of users and ensure the accountability of institutions that implement these technologies. Issues such as the protection of personal data and information security also emerge as critical challenges, requiring a multidisciplinary approach to address the complexities associated with AI governance.

Sustainable governance is also a crucial aspect in integrating AI into academic repositories. [Strecker et al. \(2023\)](#) argue that long-term data preservation requires collaborative governance models that prioritize infrastructure resilience and adaptability to technological change. This view is complemented by [Azeroual \(2021\)](#), who reinforces the importance of data quality as an essential pillar for the ongoing integrity of digital collections. In this context, [Bossaller and Million \(2023\)](#) highlight the challenges associated with data management and the need to integrate modern solutions without compromising the reliability of information already stored.

The future of AI in academic repositories presents several promising directions for future research. One priority area is the development of modular and adaptable AI tools that can be customized according to institutional and regional contexts. [Bamgbose et al. \(2024\)](#) suggest that these solutions can democratize the use of AI, enabling institutions with limited resources to implement advanced digital preservation practices. This approach could be explored in future research that evaluates how these modular tools can be designed to maximize their impact in different contexts.

Another promising direction is the construction of global networks of academic repositories, which facilitate interoperability and promote the sharing of good practices. [Moyo and Bangani \(2023\)](#) argue that international partnerships are essential to connect repositories in a global ecosystem, expanding the reach and relevance of collections. Furthermore, [Frank and Wylie \(2023\)](#) highlight the role of semantic enrichment as a facilitator for interdisciplinary connections, allowing data stored in academic repositories to be used more broadly and impactfully.

Finally, ethics must remain at the center of discussions about AI in academic repositories. [Shal et al. \(2024\)](#) emphasize that organizational leaders play a crucial role in creating policies that promote transparency, inclusion and social responsibility. [Melero et al. \(2023\)](#)

complement this perspective by pointing out that the dissemination of scientific knowledge must be accompanied by practices that ensure equity in access to and use of information. The ethical and sustainable integration of AI is essential to ensure that the benefits of these technologies are widely distributed and that their impact is positive for science and society as a whole.

AI has already proven to be an indispensable tool for transforming academic repositories, but its full impact is yet to come. To realize the full potential of these technologies, it is essential that their implementation is guided by ethical principles, infrastructure investments and collaborative efforts. By balancing technological innovation with values of inclusion and sustainability, AI can establish itself as a catalyst for scientific advancement, fostering a future in which academic knowledge is accessible and impactful for all. This vision demands a continued commitment to research and development, exploring new approaches that expand the horizons of digital preservation and connect human knowledge on a global scale.

Conclusion

The reflections presented in this article highlight that the application of AI in HEL goes beyond the simple automation of processes, representing a significant opportunity to reimagine the future of digital preservation. Faced with the challenges imposed by the exponential growth of data, rapid technological obsolescence and the scarcity of specialized resources, AI emerges as a strategic solution capable of profoundly transforming knowledge management practices, promoting efficiency, accessibility and sustainability.

A promising direction for advancement in this area lies in the development of AI tools that combine technological innovation with accessibility and inclusion. These solutions should be designed to meet the specific needs of different institutions, considering infrastructure and resource constraints. The personalization and adaptability of these technologies not only democratize access to innovations but also ensure that their benefits are distributed equitably.

Building global networks of interconnected repositories based on common standards is another key aspect of maximizing the impact of digital preservation. Data harmonization and semantic enrichment are essential to foster interoperability between systems, facilitating global collaborations and expanding the reach and relevance of scholarly collections. These initiatives strengthen libraries' ability to connect researchers and institutions in an integrated knowledge network.

Furthermore, successful AI integration will depend on clear institutional policies and organizational leadership committed to transparency, inclusion and accountability in the use of these technologies. It is essential that risks associated with the application of AI, such as algorithmic biases and inadvertent exclusions, are addressed through robust ethical guidelines and effective regulations. Ensuring trust in the technologies implemented is essential for AI to fulfill its potential as a catalyst for innovation in the field of digital preservation.

The sustainability of scholarly repositories requires a collaborative and holistic approach that incorporates not only technological advances but also strategies that strengthen infrastructure and promote inclusive and ethical practices. Digital preservation, supported by AI, offers a clear path to ensuring that scientific knowledge remains accessible and impactful across generations.

This review offers several contributions:

- It synthesizes current knowledge on AI applications in digital preservation within HEL, identifying gaps and opportunities.

- It proposes an integrative conceptual model (Figure 2) that bridges AI technology with digital preservation processes.
- It provides practical insights for library professionals and policy-makers by outlining strategies for sustainable and ethical AI integration in repositories.

It is concluded that the use of AI in HEL is more than a solution to contemporary challenges; it is a point of reflection that redefines the way knowledge is preserved and disseminated. By integrating technological advances with ethical values and collaborative practices, libraries can consolidate themselves as transformative agents in the academic ecosystem, leading the way toward a future where knowledge is universally accessible, relevant and inclusive.

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Appendix

The screening and selection process of the studies included in this review is illustrated in [Appendix](#), following the PRISMA flow diagram framework.

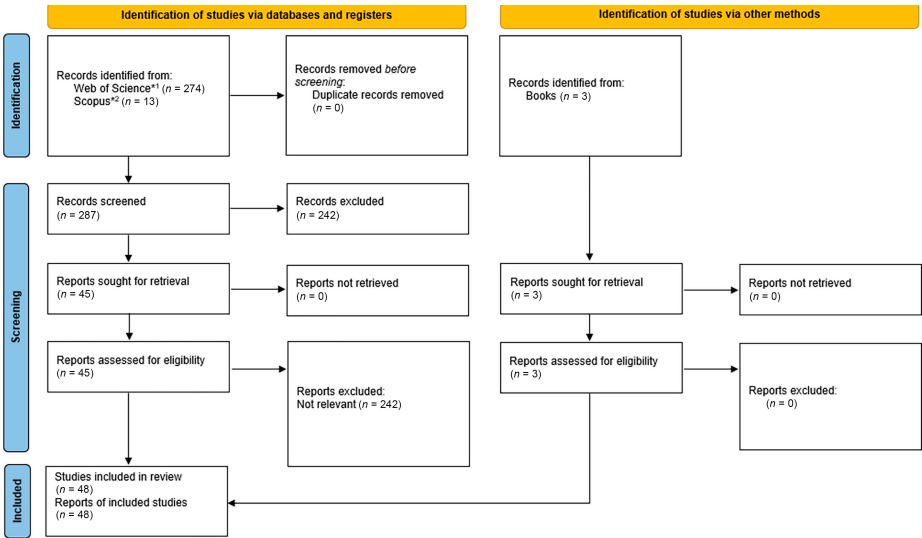


Figure A1. Prisma Flux Diagram

Source: Elaborated based on Prisma 2020 Flux Diagram

*¹ The following refining techniques were used:

- Timeline (2020–2024+2025).
- Document type (*article*)
- Open access
- Research area (information science, library science)

*² The following refining techniques were used:

- Timeline (2020–2024+2025).
- Document type (*article*).

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